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# Confidence-Aware RAG Pipeline with Hallucination Detection

*Answer only when the sources prove it, and say why when they don't.*

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Cornerstone Project

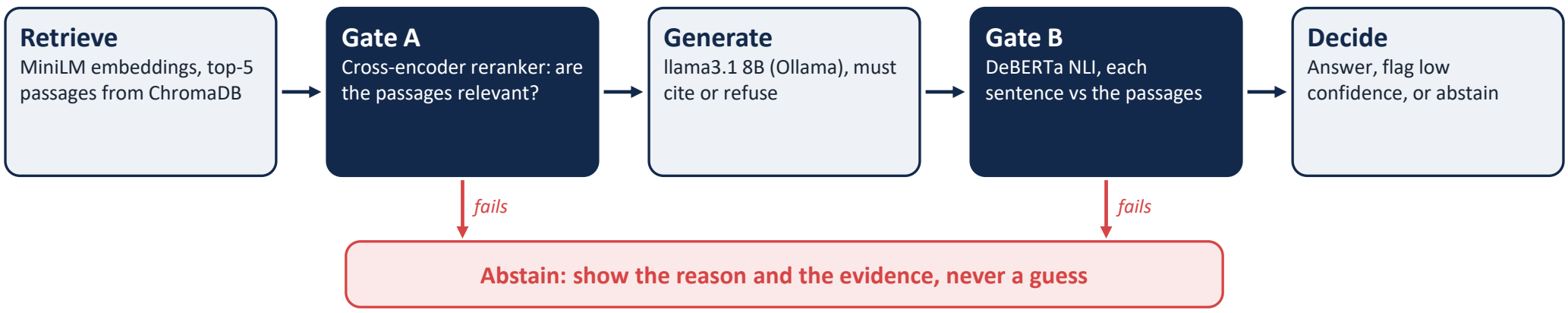
## The problem

Retrieval-augmented chatbots still hallucinate. They answer in the same confident tone whether or not the retrieved passages support the answer, so three failures reach users: questions the documents cannot answer, invented details, and statements that contradict the source. A confident wrong answer costs more than an honest "I don't know".

## Objectives

- Check passage relevance before generating (Gate A)
- Verify each answer sentence against the sources (Gate B)
- Abstain with a stated reason when confidence is low
- Compare against plain RAG on SQuAD 2.0 and HotpotQA

## Methodology: two gates, one decision




**Combined confidence** is the weighted harmonic mean of Gate A (0.4) and Gate B (0.6): one weak gate drags it down, so good retrieval cannot hide an unsupported answer. A contradicted sentence caps Gate B at 0.15.

## Key features

### Refuses before guessing

Out-of-scope questions stop at Gate A, before the language model ever runs.

### Sentence-level evidence

Every sentence labelled supported, baseless or contradicted (RAGTruth).

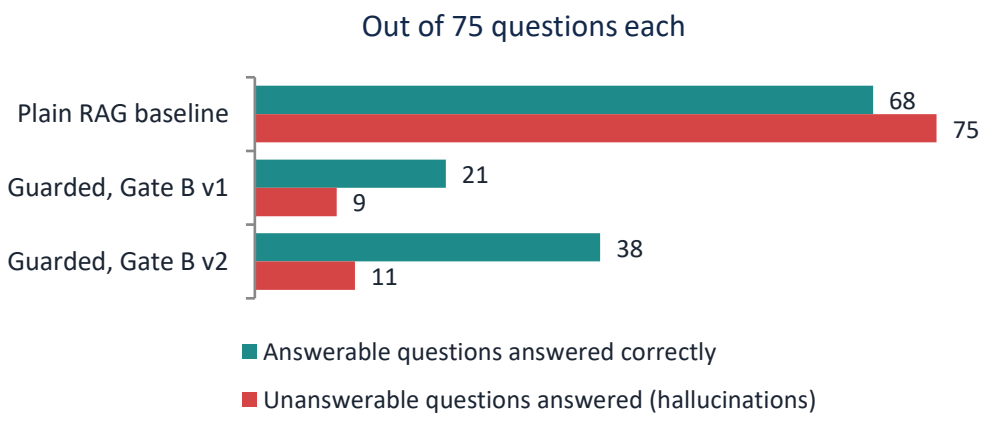
### Gate B v2

Claims checked against passages and 3-sentence windows: fewer false rejections.

### App and monitoring

Streamlit app shows every gate decision; a dashboard tracks trends and latency.

## Results: 150 SQuAD 2.0 questions, fresh end-to-end runs




**75 → 11**
hallucinated answers (of 75): plain RAG vs guarded

**97%**
of its answers to answerable questions are correct (38 of 39)

**71% → 48%**
over-abstention after Gate B v2 (paired test, $p < 0.001$)

**31% → 66%**
HotpotQA full supporting-fact recall with chunking

### Applications

- Course and campus assistants
- Policy and manual Q&A
- Healthcare, legal, finance lookups

### Future scope

- Stronger NLI model (DeBERTa-large)
- Prompt-only control arm
- Multi-passage entailment for multi-hop

### Built with

- Python · LangChain · ChromaDB
- DeBERTa-v3 NLI · MiniLM reranker
- Ollama llama3.1 8B · Streamlit